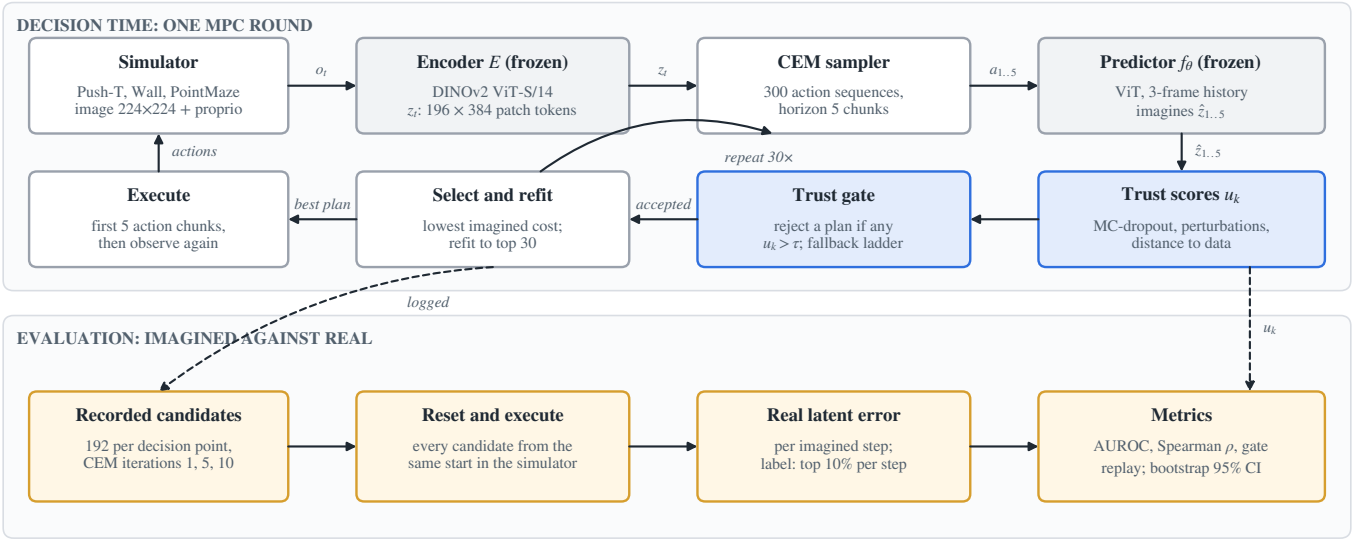
DECISION TIME: ONE MPC ROUND
Simulator
Push-T, Wall, PointMaze
image 224×224 + proprio
o_t
Encoder E (frozen)
DINOv2 ViT-S/14
z_t: 196 × 384 patch tokens
z_t
CEM sampler
300 action sequences,
horizon 5 chunks
$a_{1..5}$
Predictor f_θ (frozen)
ViT, 3-frame history
imagines $\hat{z}_{1..5}$
$\hat{z}_{1..5}$
Trust scores u_k
MC-dropout, perturbations,
distance to data
repeat 30×
Trust gate
reject a plan if any
$u_k > \tau$; fallback ladder
accepted
Select and refit
lowest imagined cost;
refit to top 30
best plan
Execute
first 5 action chunks,
then observe again
actions
logged
u_k
EVALUATION: IMAGINED AGAINST REAL
Recorded candidates
192 per decision point,
CEM iterations 1, 5, 10
Reset and execute
every candidate from the
same start in the simulator
Real latent error
per imagined step;
label: top 10% per step
Metrics
AUROC, Spearman ρ, gate
replay; bootstrap 95% CI